

CHAPTER TWO

NATURE OF MUNICIPAL WASTEWATERS

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1. WASTEWATER CHARACTERIZATION

In so far as it affects biological treatment, municipal waste flows can be characterized chemically and physically.

Chemical characterization comprises identification and measurement of: (a) carbonaceous and nitrogenous constituents in their various fractions, soluble and particulate, biodegradable and unbiodegradable; and (b) inorganic constituents principally total alkalinity (Alkalinity), total acidity, pH and phosphorus. Inorganic constituents such as calcium, sodium, magnesium, chloride and sulphate usually are of minor importance with the exception of calcium which, in certain circumstances, can be of significance in phosphorus removal by calcium phosphate precipitation.

Physical characterization comprises separation of the wastewater into dissolved, suspended and settleable constituents.

2. CHEMICAL CHARACTERIZATION

2.1 Carbonaceous materials

Characterization of the carbonaceous material in the influent is done via the Chemical Oxygen Demand (COD)

test. For activated sludge process design it is necessary to identify and know the magnitudes of the various fractions of the influent COD as these significantly affect the response of the process. Research at the University of Cape Town has indicated that the subdivisions shown diagrammatically in Figure 2.1 are essential for accurate description of the behaviour of the process.

2.1.1 Biodegradable and unbiodegradable fractions

From Figure 2.1 the first major subdivision is into biodegradable COD (S_{bi}) and unbiodegradable COD (S_{ui}) fractions. This division is important for the following reasons: In design, knowing biodegradable COD concentration and the flow per day gives the biodegradable COD load on the plant; knowing the mass of biodegradable COD load per day and selecting a sludge age, the daily carbonaceous oxygen requirements and the active mass in the process can be estimated from the kinetic equations governing the process (Eq 4.10, Chapter 4). Conversely the fraction of the total influent COD which is biodegradable can be estimated from the *carbonaceous* oxygen consumption rate in a steady state laboratory scale completely mixed activated sludge process run at one or more sludge ages. The mass of S_{bi} is then estimated by inserting the measured oxygen demand into the steady state equa-

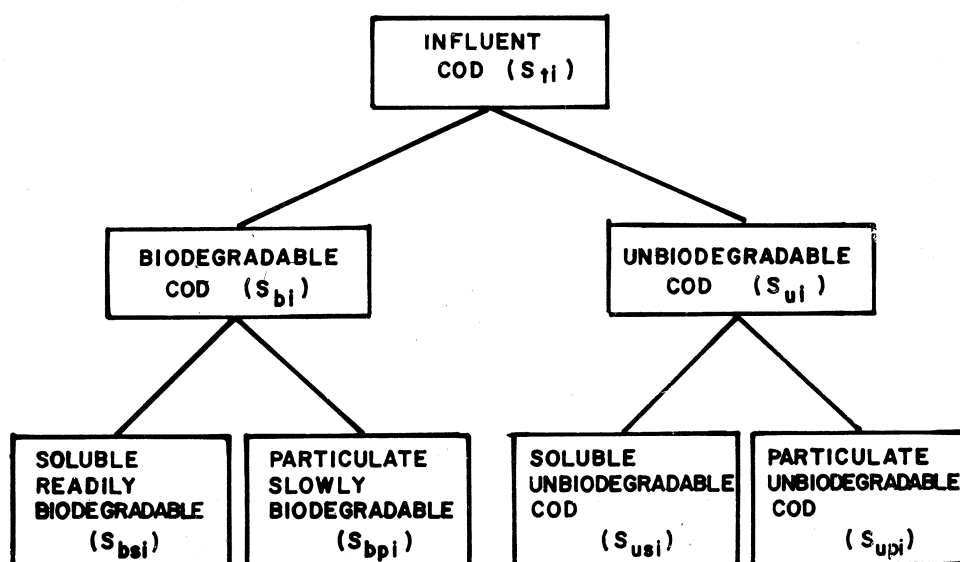


FIGURE 2.1: Division of the total influent COD in municipal wastewaters into its various constituent fractions

tion for the carbonaceous oxygen demand and by trial and error finding the value of S_{bi} that balances the equation, (Eq 4.15, Chapter 4). This approach will be adequate for developing information for design. It presumes that the various constants in the kinetic equations, such as the specific yield constant Y_h and the specific endogenous mass* rate constant b_h have been correctly evaluated from research studies, (Marais and Ekama, 1976). It presumes also that all the biodegradable COD has been metabolized, a presumption justified from research findings if the sludge age is greater than 3 days at 20°C, (Dold, Ekama and Marais, 1980). If nitrification occurs in the process this either has to be allowed for when determining the carbonaceous oxygen demand from the measured oxygen demand (see Chapter 5) or, nitrification must be suppressed by addition of a nitrification inhibitor to the process, for example by addition of thio-urea at intervals to the process.

2.1.2 Unbiodegradable fractions

The next major subdivision is to divide each of the unbiodegradable and biodegradable COD concentrations, S_{ui} and S_{bi} respectively, into their relevant subconcentrations.

Consider first the unbiodegradable COD concentration, S_{ui} . This concentration consists of two fractions, an unbiodegradable soluble COD, S_{usi} and an unbiodegradable particulate COD, S_{upi} . Both fractions are hypothesised to be unaffected by biological action so that at steady state the masses that enter equal the masses that leave the process. The S_{usi} passes out in the liquid effluent discharge. The S_{upi} is enmeshed in the sludge, accumulates in the system to a mass equal to that entering in the influent per day, multiplied by the sludge age, but eventually at steady state the masses entering and leaving the system are the same. The mass leaves the system via the daily sludge mass wasted from the plant. Thus the S_{upi} has the principal effect of increasing the mixed liquor concentration.

Usually it is more convenient to express the unbiodegradable particulate influent fraction not in terms of its COD but in terms of its influent volatile solids concentration, X_{ij} . This is readily accomplished by noting in Chapter 1 that the COD and volatile solids are defined as follows:

$$X_{ij} = S_{upi}/f_{cv}$$

where f_{cv} = COD/VSS ratio = 1,48 mgCOD/mgVSS.

Measurement of S_{usi} is relatively simple; a laboratory scale unit is run at say two different sludge ages both greater than 3 days. The COD of the centrifuged effluent or the filtrate through a fine filter, estimates S_{usi} . The balance of the influent COD, after subtracting S_{bi} and S_{ui} from the total influent COD, S_{ti} , is the unbiodegradable particulate COD, S_{upi} , i.e.

$$S_{upi} = S_{ti} - S_{bi} - S_{usi}$$

The reasonableness of the S_{upi} estimate is checked by running a laboratory scale unit at a long sludge age, say, 15 days or longer, calculating the total volatile solids concentration using the input values for S_{bi} and S_{upi} (i.e. X_{ij}) (Eq 4.13, Chapter 4) and checking this value against the volatile mass observed. A long sludge age is selected because the total volatile mass of sludge becomes more sensitive to X_{ij} as the sludge age increases. Overall the S_{upi} , S_{ui} and S_{bi} must give rise to consistency at the different sludge ages. Once this consistency is judged adequate the COD fractions are acceptable for design.

2.1.3 Biodegradable fractions

The data obtained above are all on steady state aerobic activated sludge units. Subdivision of S_{bi} into readily biodegradable soluble COD, S_{bsi} , and slowly biodegradable particulate COD, S_{bpi} , (Fig. 2.1) can be obtained only by dynamic loading of the units, for the following reasons: The fraction S_{bsi} is rapidly taken up by the sludge in a matter of minutes and metabolized, giving rise to a high unit rate of oxygen demand for synthesis. (The kinetics of utilization are in accordance with Monod's equation). The fraction S_{bpi} requires to be adsorbed and stored on the organism, broken down to simpler chemical units by extracellular enzymes and then absorbed and metabolized by the organism. The extracellular breakdown is slow and forms the limiting rate in the synthesis reaction, the rate being about 1/7 to 1/10 that for the readily biodegradable COD, S_{bsi} . (The kinetics for synthesis of particulate COD is formulated via Levenspiel's active site surface reaction theory). Under steady state (i.e. constant flow and load conditions) the stored S_{bp} is negligibly small at sludge ages greater than about 2 to 3 days. However at short sludge ages (2-3 days) under cyclic flow conditions under square wave loading, 12 h on, 12 h off, during the loading period the rate of S_{bp} adsorption and storage is so large that the organism cannot deplete it concurrently, even at the maximum utilization rate. As a consequence, after the loading period terminates the utilization rate of the stored S_{bp} continues at the same maximum rate for a while, and then slowly declines. This behaviour allows the readily biodegradable COD to be determined; measure the oxygen utilization rate for a period before loading terminates and a period after, the difference in the rates defines the rate due to the S_{bsi} . Details of the test procedure are given in Appendix A2.

The importance of the S_{bsi} fraction will become evident when denitrification and biological excess phosphorus removal are discussed in Chapters 6 and 7 respectively.

2.1.4 Analytic formulation

For analysis the relationship indicated in Figure 2.1 can be expressed as follows:

* Y_h = 0,45 mgVASS/mgCOD utilized, independent of temperature
 b_h = 0,24 mgVASS/mgVASS/day at 20°C and
 b_{hT} = 0,24 (1,029)^(T-20) at temperature T°C, see Chapter 4.

Biodegradable and unbiodegradable COD fractions:

$$S_{ti} = S_{ui} + S_{bi} \quad (2.1)$$

where

S_{ti} = total influent COD concentration (mgCOD/l)

S_{ui} = unbiodegradable COD concentration (mgCOD/l)

S_{bi} = biodegradable COD concentration (mgCOD/l)

Each of the two fractions on the right hand side of Eq 2.1 is again subdivided, Figure 2.1.

Unbiodegradable COD fractions:

The unbiodegradable COD concentration consists of two components, soluble and particulate, i.e.

$$S_{ui} = S_{usi} + S_{upi} \quad (2.2)$$

where

S_{usi} = unbiodegradable soluble COD concentration (mgCOD/l)

S_{upi} = unbiodegradable particulate COD concentration (mgCOD/l).

It is convenient to express S_{usi} and S_{upi} empirically in terms of the total COD concentration S_{ti} , i.e.

$$S_{usi} = f_{us} S_{ti} \quad (2.3)$$

$$S_{upi} = f_{up} S_{ti} \quad (2.4)$$

where

f_{us} = unbiodegradable soluble COD fraction with respect to the total COD (mgCOD/mgCOD)

f_{up} = unbiodegradable particulate COD fraction with respect to the total COD (mgCOD/mgCOD).

Hence from Eq (2.2)

$$S_{ui} = (f_{us} + f_{up}) S_{ti} \quad (2.5)$$

The particulate unbiodegradable COD concentration may be expressed also in terms of volatile solids, X_{ij} :

$$\begin{aligned} X_{ij} &= S_{upi} / f_{cv} \\ &= f_{up} \cdot S_{ti} / f_{cv} \end{aligned} \quad (2.6)$$

where

X_{ij} = unbiodegradable particulate volatile solids concentration in the influent (mgVSS/l)

f_{cv} = COD to VSS ratio of the solids
= 1.48 mgCOD/mgVSS.

Biodegradable COD fractions:

The biodegradable COD concentration is found from Eq 2.1 as follows:

$$S_{bi} = S_{ti} - S_{ui}$$

and from Eq 2.5

$$\begin{aligned} &= S_{ti} - S_{ti} (f_{up} + f_{us}) \\ &= S_{ti} (1 - f_{up} - f_{us}) \end{aligned} \quad (2.7)$$

From Figure 2.1 the *biodegradable* COD, S_{bi} , is divided into *readily biodegradable soluble* COD, S_{bsi} , and *slowly biodegradable particulate* COD, S_{bpi} . Each can be expressed in terms of S_{bi} as follows:

$$S_{bsi} = f_{bs} S_{bi} \text{ and} \quad (2.8a)$$

$$S_{bpi} = (1 - f_{bs}) S_{bi} \quad (2.9a)$$

where

f_{bs} = readily biodegradable COD fraction with respect to the *biodegradable* COD.

The readily biodegradable COD can also be expressed in terms of the *total* COD, S_{ti} , i.e. substituting for S_{bi} from Eq 2.7 in Eq 2.8 yields

$$S_{bsi} = f_{bs} (1 - f_{up} - f_{us}) S_{ti} \quad (2.8b)$$

$$= f_{ts} S_{ti} \quad (2.9b)$$

where

f_{ts} = readily biodegradable COD fraction with respect to the total COD.

2.2 Nitrogenous materials

Characterization of the nitrogenous material in the influent is with the Total Kjeldahl Nitrogen (TKN) and Kjeldahl free and saline ammonia tests. As for the carbonaceous material, the nitrogenous material also is subdivided into different fractions but in a different fashion from that for the carbonaceous material. The subdivision is shown in Figure 2.2.

In certain wastewaters nitrate and nitrite might be present; the TKN test does not include these. The vast majority of municipal effluents will not contain nitrate or nitrite because in most sewerage systems the wastewater will be in an unoxxygenated state and any nitrate entering the system is likely to be denitrified before it reaches the wastewater treatment plant.

2.2.1 Free and saline and proteinaceous fractions

The free and saline ammonia fraction, N_{ai} , is determined by a test of this name. The proteinaceous or organic fraction is determined by difference between the TKN, N_{ti} , and free and saline ammonia values, N_{ai} .

The free and saline ammonia is immediately available for incorporation into the bacterial protoplasm and for conversion to nitrite or nitrate, if the process is appropriately designed.

The proteinaceous fraction, from Figure 2.2, consists of a number of sub-fractions, *unbiodegradable soluble* and *unbiodegradable particulate organic nitrogen* fractions, N_{ui} and N_{pi} respectively, and a *biodegradable organic nitrogen* fraction, N_{oi} . The unbiodegradable soluble fraction passes unaffected through the system and is discharged in the effluent similar to the soluble unbiodegradable COD. The unbiodegradable particulate organic nitrogen by implication must be part of the unbiodegradable particulate COD (as X_{ij}), and is treated as such; hence, this fraction leaves the process via the sludge wasted daily. The

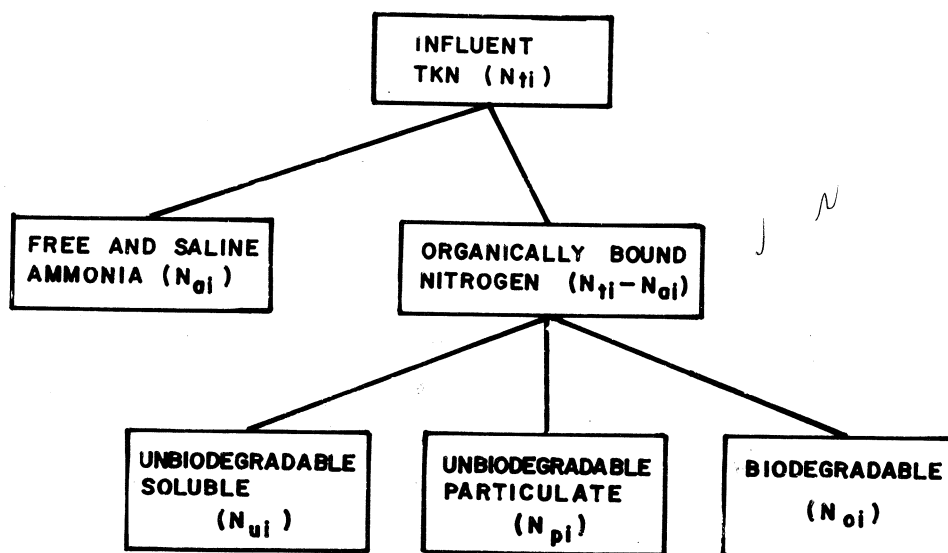


FIGURE 2.2: Division of the total influent TKN in municipal wastewater into its various constituent fractions

biodegradable organic nitrogen is broken down by the heterotrophs to free and saline ammonia which with its influent counterpart participates in any further biologically mediated reactions. The breakdown occurs very readily and normally it can be expected to be virtually complete for all sludge ages greater than about 3 days.

With regard to measurement of these fractions, the free and saline ammonia (N_{ai}) is measured directly by the test bearing this name; the total organic nitrogen is found from the difference between the TKN and free and saline ammonia test concentrations. Difficulties in quantification arise when subdividing the total organic nitrogen into its biodegradable and unbiodegradable fractions, N_{oi} , N_{ui} and N_{pi} , (see Figure 2.2). N_{ui} and N_{pi} are postulated by association to their counterparts in the COD sub-division. Estimates of their magnitudes can be obtained only by comparing the observed response of laboratory scale processes with that predicted by the theoretical model incorporating these fractions. From a large number of such comparisons on processes operated within the range of sludge ages from 2 to 30 days, under cyclic and steady states in the temperature range 12 to 25°C, it would appear that N_{ui} and N_{pi} are necessary in the model to obtain good and consistent correlation between observation and prediction. However, the fractions are small and it is not necessary therefore to have accurate estimates. Concurrently N_{pi} is expressed in terms of the unbiodegradable particulate volatile solids in the influent (X_{ij}) i.e. N_{pi} is 10% of X_{ij} (see Eq 2.13): from experimental data on the TKN/VSS ratio of the MLVSS in the biological reactor, using the same influent wastewater source it was found that the TKN/VSS ratio remained the same irrespective of sludge age from 3 to 30 days despite the fact that the various VSS fractions change with sludge age. Hence it is reasonable to accept that the TKN/VSS ratio of the inert organic solids is equal to that of the other VSS fractions at 0,10 mgN/mgVSS.

TABLE 2.1 APPROXIMATE AVERAGE MUNICIPAL WASTEWATER CHARACTERISTICS FOR RAW AND SETTLED WASTEWATERS IN SOUTH AFRICA

Wastewater Characteristic	Raw	Settled
1. Influent COD (mgCOD/l)	500-800	300-600
2. Influent BOD ₅ (mgBOD/l)	250-400	150-300
3. Influent TKN (mgN/l)	35-80	30-70
4. Influent Phosphorus (mgP/l)	8-18	6-15
5. Total Suspended Solids (mg/l)	270-450	150-300
6. Settleable Solids in (mg/l)	150-350	0-50
in (ml/l)**	6-14	0-2
7. Non-Settleable Solids (mg/l)	100-300	100-300
8. TKN/COD ratio (mgN/mgCOD)	0,07-0,10	0,09-0,12*
9. Total P/COD ratio (mgP/mgCOD)	0,015-0,025	0,020-0,030*
10. Unbiodegradable particulate COD fraction (f_{up})	0,07-0,20	0,00-0,10
11. COD/VSS ratio of unbiodegradable particulate COD (f_{cv}) (mgCOD/mgVSS)	1,45-1,50 ave. 1,48	1,45-1,50 ave. 1,48
12. TKN/VSS ratio of unbiodegradable particulate COD (f_n) (mgN/mgVSS)	0,09-0,12 ave. 0,10	0,09-0,12 ave. 0,10
13. Unbiodegradable soluble COD fraction (f_{us})	0,04-0,10	0,05-0,20*
14. Fraction biodegradable COD (S_{bi}/S_{ti})	0,75-0,85	0,80-0,95
15. Readily biodegradable COD with respect to total (f_{is})	0,08-0,25	0,10-0,35*
16. Maximum specific growth rate of the nitrifiers (/d)	0,20-0,70	0,20-0,70
17. Ammonia/TKN fraction (f_{na})	0,60-0,80	0,70-0,90*
18. Unbiodegradable soluble TKN fraction (f_{nu})	0,00-0,04	0,00-0,05*
19. Minimum temperature (°C)	10-15	10-15
20. Maximum temperature (°C)	20-30	20-30
21. Alkalinity (mg/l as CaCO ₃)	200-300	200-300

*Note that the characteristics of settled wastewater should be selected in conformity with the characteristics of the original raw wastewater and primary settling tank behaviour (see Chapter 4, Section 12.1).

**Volume settled in a 1 l Imhoff cone in 2 hours.

Furthermore the unbiodegradable soluble organic nitrogen concentration (N_{ui}) apparently is very low, between 0 and 5% of the influent TKN for raw wastewaters. Later it will be shown (Chapter 5) that the N_{ui} concentration plays a very minor role in design. The suggested fractional values are listed in Table 2.1.

2.2.2 Analytical formulation

The total TKN, N_{ti} is divided into its constituents as follows, (see Figure 2.2):

$$N_{ti} = N_{ai} + N_{ui} + N_{pi} + N_{oi} \quad (2.10)$$

where

N_{ti} = total influent TKN (mgN/l)

N_{ai} = influent ammonia (mgN/l)

N_{ui} = soluble unbiodegradable organic nitrogen in the influent (mgN/l)

N_{pi} = particulate unbiodegradable organic nitrogen in the influent (mgN/l)

N_{oi} = biodegradable organic nitrogen in the influent (mgN/l)

It is convenient to express some of the fractions of nitrogen in terms of the total TKN concentrations

$$N_{ai} = f_{na} N_{ti} \quad (2.11)$$

$$N_{ui} = f_{nu} N_{ti} \quad (2.12)$$

where

f_{na} = ammonia fraction of the influent TKN (mgN/mgN)

f_{nu} = unbiodegradable soluble organic nitrogen fraction (mgN/mgN)

The unbiodegradable organic nitrogen in the particulate material, N_{pi} , is best expressed in terms of the unbiodegradable influent volatile material, X_{ii} , or in terms of its COD counterpart, i.e.

$$N_{pi} = f_n S_{upi}/f_{cv} = f_n X_{ii} \quad (2.13)$$

where

f_n = nitrogen fraction of the influent biodegradable volatile particulate material ≈ 0.10 mgN/mg X_{ii}

From Eqs 2.10 to 2.13, the biodegradable organic nitrogen (N_{oi}) can be found by subtracting N_{ui} , N_{ai} and N_{pi} from N_{ti}

$$N_{oi} = N_{ti} (1 - f_{na} - f_{nu}) - f_n f_{up} S_{ti}/f_{cv} \quad (2.14)$$

The significance of the various influent TKN fractions on the design of nutrient removal activated sludge processes is discussed in Chapter 3, Section 3.2 and in Chapter 5, Section 5.1.

2.2.3 Specific growth rate of nitrifiers

Experimental observations on a number of different municipal wastewater flows have indicated that the

maximum specific growth rate constant of the nitrifiers, μ_{nm} , can differ greatly in value and appears to be specific to each waste flow. The magnitude of μ_{nm} can have a significant influence on the design of nutrient removal processes, see Chapters 3, 4, 5 and 6. Therefore, for optimal design it is most desirable to have an estimate of μ_{nm} . This estimate is obtained by running a completely mixed single reactor at a sludge age of about 8 days, at 20°C imposing a sequence of aerobic and anoxic periods on it and measuring the rate of nitrate increase (or the rate of Alkalinity decrease) during the aerobic period. The data thus obtained can be analysed manually to give the μ_{nm} at 20°C, i.e. μ_{nm20} . By applying the well established relationship between μ_{nm} and temperature the value at any temperature can be determined. Details of the test procedure and analysis of the data are given in Appendix 3.

2.3 Phosphorus

The total phosphorus concentration in municipal wastewaters consists mainly of two fractions, a soluble orthophosphate (PO_4^{3-}) fraction and an organically bound phosphorus fraction which may be soluble or particulate in form. In both settled and raw municipal wastewaters, the orthophosphate fraction predominates, ranging between 70 to 90 per cent of the total phosphorus.

The orthophosphate concentration* can be measured by a colorimetric test of that name. However, this measure records only the orthophosphate concentration of the influent, which may underestimate the total phosphorus in the influent by about 10 to 20 per cent or more. The organically bound phosphorus is converted to orthophosphate in the activated sludge process so that the orthophosphate concentration in the influent underestimates the phosphorus to be removed in the process. For example, if the orthophosphate concentration is 8 mgP/l and the total-P is 10 mgP/l then for an effluent standard of 1 mgP/l orthophosphate, if the plant is designed to remove 7 mgP/l on the basis of the ortho-P influent concentration, the effluent actually will contain 3 mgP/l ortho-P. The process should be designed to remove 9 mgP/l on the basis of the total-P influent concentration. Consequently, for the influent the only reliable test is the total-P test. For the effluent, the filtered ortho-P concentration very closely approximates the filtered total-P concentration and this ortho-P test would appear to be adequate. However, in performing the ortho-P test, a problem sometimes is encountered in that the natural colour in some wastewaters interferes with the colorimetric test giving rise to spurious results.**

3. PHYSICAL CHARACTERIZATION

Physical characterization of the influent exists mainly in estimating the expected performance of the primary settling tank. The function of the settling tank is to separate in some degree the settleable particulate

*All phosphorus and phosphate concentrations are measured relative to the phosphorus atom i.e. mgP/l.

**This is a problem encountered in treated effluents from the Western Cape, but apparently is absent from treated effluents of Johannesburg and Pretoria.

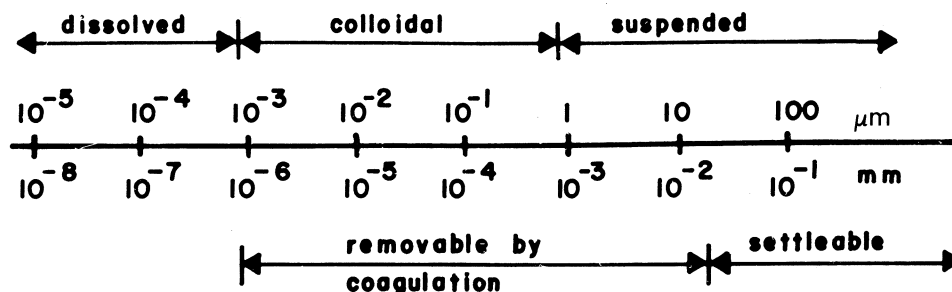


FIGURE 2.3: Particle size range and classification

material from the influent flow to give a supernatant overflow having reduced values of COD, TKN and phosphorus.

Physical characterization basically is made on the relative size of the particles in the wasteflow. Here three broad divisions are recognized, *dissolved*, *colloidal* and *suspended solids*, see Figure 2.3. Generally material of particle size smaller than 10⁻³ to 1 μm is accepted as colloidal and material with particle size larger than 1 μm as suspended. To obtain estimates of the suspended material pore size membrane filters other than 1 μm are in use which can lead to significantly different results so that when reporting or interpreting data, careful note should be made of the pore size of the filter membranes (and papers) on which the data are based.

The fraction of *suspended solids* (i.e. retained on a 1 μm filter) that can be removed by gravity settling is called the *settleable solids* and usually ranges in equivalent pore size from 10 to 50 μm depending on particle density. However, a more practical procedure to estimate the mass of settleable solids in a sample of wastewater, is to use the Imhoff cone: A one litre Imhoff cone is filled with a sample and allowed to settle for two hours. The volume of solid material settled into the bottom of the cone gives a measure (in ml/l) of the quantity of material that can be removed by gravity sedimentation. From a large number of tests it has been found that 1 ml/l of settleable solids is approximately equivalent to a dry mass* of 25 mg/l; with this equivalence it is possible to obtain an estimate of the mass of settleable solids using the Imhoff test.

The settleable solids can be measured either as *total solids* where total solids include both the organic and inorganic fractions, or, as *volatile or organic solids* where the volatile solid is the difference between the total solids and the ash remaining after the total solids are incinerated in an oven at 600°C for half an hour. At this temperature most inorganic (mineral) salts, except magnesium carbonate, do not decompose or volatilize. The ratio Volatile Solids:Total Solids of the settleable solids from municipal wasteflows is about 0,75 i.e. 75% volatile, 25% mineral so that Volatile Solids $\cong$ 0,75 Total Solids. The COD associated with the volatile solids can

be estimated by multiplying the volatile solid mass by 1,48 i.e. COD $\cong$ 1,48 Volatile Solids.

The Imhoff cone test serves as a means whereby the percentage removal of Total Phosphorus and TKN in the primary settling tank can be estimated. This is done by determining total-P and TKN concentrations in the waste sample before settling and in the supernatant after settling in the cone.

In a primary settling tank the degree of COD and settleable solids removed by gravity sedimentation depends on the upflow velocity (overflow rate) and retention time. For relatively deep tanks (>3 m) with mean retention times of 2 to 3 h and mean overflow rates of 1,5 to 2 m/h between 50 and 80% of the settleable solids (as given by the Imhoff cone test) and 30 to 50% (mean of 40%) of the total COD will be removed. Increasing the retention time above 3 h does not improve the removals significantly. However, lowering the retention time to say one hour (equivalent to 3 m/h mean overflow rate) reduces the removal to about half of that at 3 h retention time.

With regard to the TKN and P removal in primary tanks this can vary depending on state of the waste flow. It seems that in fresh wastewater a larger fraction of the TKN and P may be bound to the settleable solids than if the wastewater is old and solubilization of the TKN and total P fractions has occurred. Unfortunately insufficient information on these aspects is available to lay down criteria for design. For design it would seem preferable to assume that solubilization does occur so that only a minor fraction of the TKN and P respectively is removed by primary sedimentation, between 15 and 25%, even though there are reported removals of up to 40%.

The discussion above indicates that with primary sedimentation a significant reduction is obtained in COD (40%) but a smaller reduction of TKN and total P (15 to 20%). This has the effect that settled wastewater has increased TKN/COD and P/COD ratios compared to the raw wastewaters. Note also that the readily biodegradable COD concentration (S_{bs}) which is completely in the dissolved form is not reduced so that the concentrations in the raw and settled flow are the same,

*Mass retained on filter paper (Whatmans No. 1) after drying for 12 hours at 105°C. Measurements taken from raw wastewater at the Cape Flats treatment works by Ekama, Marais and coworkers.

hence the fraction S_{bi}/S_{ti} increases for settled sewage. With regard to the unbiodegradable particulate COD this fraction appears to be largely removed by primary sedimentation so that settled wastewaters contain low fractions of unbiodegradable particulate COD.

4. QUANTIFICATION OF WASTEWATER CHARACTERISTICS

In the above section the various chemical and physical wastewater characteristics were defined. In this section

these characteristics are quantified to facilitate their use in design. The influence of the various characteristics on the design of an activated sludge process is discussed in Chapter 3.

The ranges of the magnitudes of the various chemical and physical characteristics of approximately normal *raw* and *settled* municipal wastewaters is such that it is difficult to give smaller ranges for these characteristics. Generally it has been found that the greater the diversity and larger the community from which the wastewater originates, the less the characteristics will vary during the life of the plant. Also

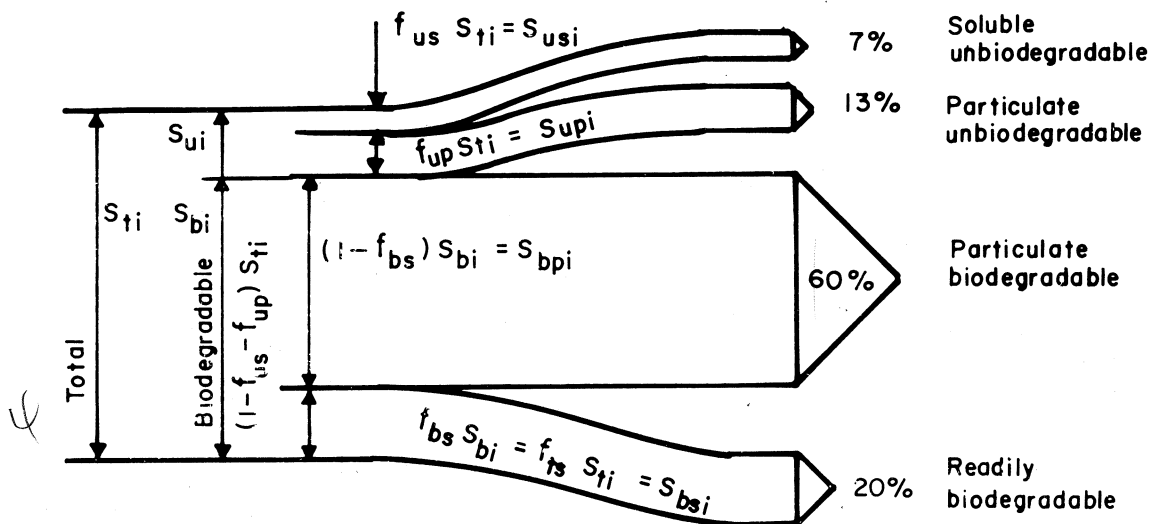
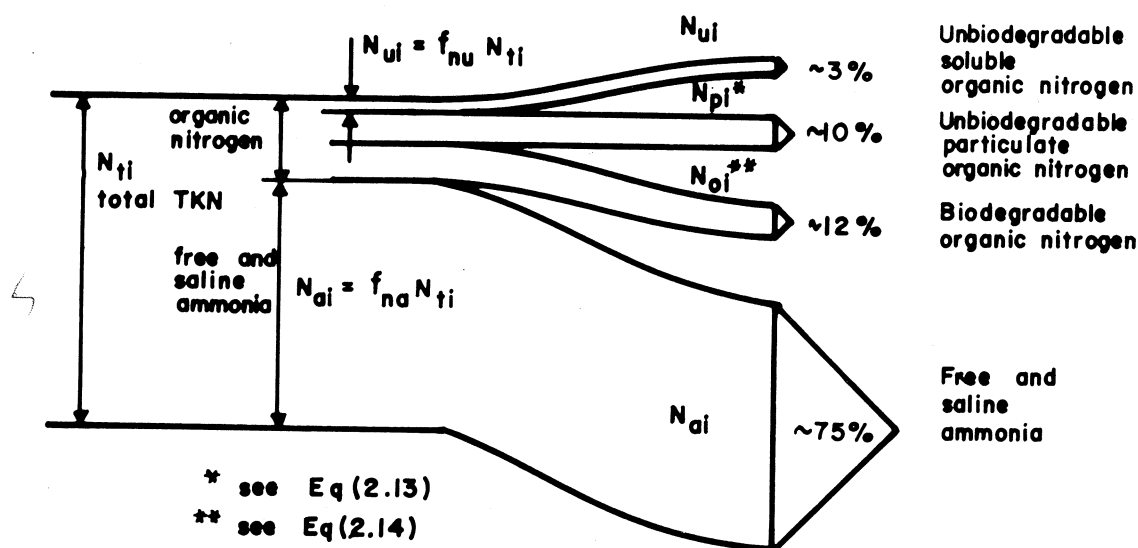


FIGURE 2.4: Diagrammatic representation of the different fractions of the total influent COD concentration. Numerical values apply to raw municipal wastewater



* see Eq (2.13)
** see Eq (2.14)

FIGURE 2.5: Diagrammatic representation of the various fractions of the influent TKN in municipal wastewater. Numerical values apply to raw municipal wastewater

the greater the diversity and larger the community, the smaller the daily variations in wastewater characteristics and flow.

A diagrammatic representation of the magnitudes of the different COD and TKN fractions in an approximately normal *raw* municipal wastewater is shown in Figures 2.4 and 2.5 respectively. With regard to the COD fractions, Figure 2.4 shows that about 7% of the total COD is soluble unbiodegradable, 13% particulate unbiodegradable, 60% particulate slowly biodegradable and 20% soluble readily biodegradable. With regard to the TKN fractions, Figure 2.5 shows that about 3% of the total TKN is soluble unbiodegradable organic

nitrogen, 10% particulate unbiodegradable organic nitrogen associated with the particulate unbiodegradable COD, 12% biodegradable organic nitrogen and 75% free and saline ammonia.

5. REFERENCES

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